

ESIT-2026 Challenge - Task 2

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What is the contamination?

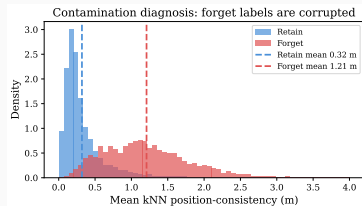
The forget labels are corrupted positions — the CSI is genuine, the stored (x, y) is wrong.

kNN-consistency (label vs. its 10 nearest *retain* neighbours, CSI-PCA space):

	Retain	Forget
mean shift	0.32 m	1.21 m
AUC(z shift)	0.936	

If the model predicts the *true* position,

$e_i \approx \|\mathbf{y}_i^{\text{true}} - \mathbf{y}_i\|_2$ — **error = corruption magnitude**, which transfers to test.



Family A — Relabel: denoise the labels

Idea: give the model a *trusted* target for each forget sample, then fine-tune. The error against the *original* label then equals the corruption.

1. For each forget sample i , find its $k = 10$ nearest **retain** neighbours in the 64-dim PCA of $|\text{CSI}|$.
2. Replace its label with their mean position: $\hat{\mathbf{y}}_i^{\text{corr}} = \frac{1}{k} \sum_{j \in \text{kNN}_{\mathcal{R}}(i)} \mathbf{p}_j$ (retain labels untouched).
3. Fine-tune on the corrected targets.

Why it works for the metric

The model learns the true position $\Rightarrow$ for a forget sample the error vs. the *original* (corrupted) label is large and **transfers exactly** to the test set.

Result: self-MIA 0.927, retain error 0.114 m, forget error 0.861 m.

Family B — Activation-trajectory divergence

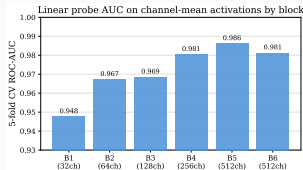
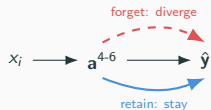
Motivation: the CNN *already* separates forget from retain internally (probe AUC **0.986** at block 5), yet parameter-space dampening (SSD) *fails* (Fisher importance entangled).

So act in activation space.

Method (vs. frozen teacher f_{θ_0}):

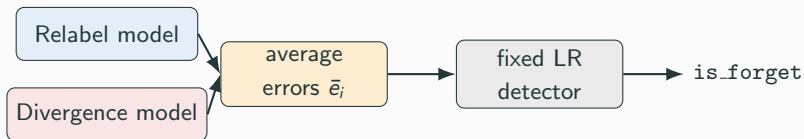
- **Diverge** forget pre-ReLU acts off-trajectory.
- **Anchor** retain post-ReLU acts to the teacher.
- + kNN forget-position loss (transferable signal).

Real representational change, not just a relabel.



The small ensemble

Why ensemble? The two families make *complementary* errors: relabel gives a clean transferable signal; divergence detaches the representation. Averaging their per-sample errors before the detector yields a **more separable** error distribution.

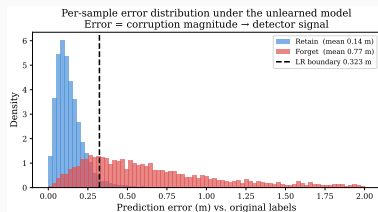


Results

Method	self-MIA	retE	fgtE	fgt%
Relabel	0.927	0.114	0.861	0.538
Divergence	0.881	0.139	0.769	0.519
Baseline	—	0.101	0.125	—

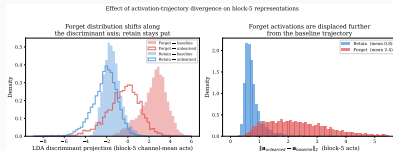
retE / fgtE = mean retain / forget error (m).

fgt% = LR test forget rate (true ≈ 0.50).



Why this is real unlearning — not just relabelling

- Divergence changes *where* the model maps forget inputs in activation space — direct evidence the internal computation changed.
- Forget activations are pushed off the teacher trajectory; retain stays anchored.
- **Ablation:** drop the kNN-position loss and the activation push alone is partly train-specific (worse transfer). The two terms are complementary: *detachment* + *transferable signal*.



Block-5 activations: **forget** displaced after unlearning, **retain** anchored.

Summary

1. **Signal**: contamination = corrupted positions; error = corruption magnitude.
2. **Family A** relabels with kNN-corrected positions (transferable signal).
3. **Family B** diverges forget activations off a frozen teacher (representational).
4. **Small ensemble** of both